

cft-anyons Lab Book
From category data to provably rigorous CFTs

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1 Frontmatter, North Star, and Claim Status

This is a reproducible research lab book. Its guiding goal — the *north star* — is a single, concrete construction problem:

Given a (unitary) fusion or modular tensor category $\mathcal{C}$, together with whatever additional data is needed (OPE coefficients / conformal data), construct a family of microscopic (lattice) models and their symmetry generators whose continuum limit *provably* yields a mathematically rigorous QFT/CFT that (i) realises the input category data and (ii) carries a full projective unitary representation of the symmetries.

The working belief is that this is achievable at least for *rational* CFTs, and perhaps for all QFTs. This is a programme, not a settled theory: every step toward the goal is a question, proposal, or hypothesis until a local source, a local derivation, or a reproducible run supports it.

1.1 The Two Success Criteria

A construction is only a candidate solution if it meets both, non-negotiably:

1. **Provability.** The continuum limit is established rigorously (an operator-algebraic / wavelet renormalisation limit with theorems, not a heuristic scaling argument).
2. **Fidelity.** The realised continuum content faithfully reproduces the input category data $\mathcal{C}$, and the symmetry representation is *full* and projective-unitary.

1.2 Claim Status

No claim is treated as established until it is marked *Source-local* or *Checked* and linked to a local source, derivation, test, or run artifact. The status labels are fixed in `CONVENTIONS.md`: *Question*, *Proposal*, *Source-local*, *Checked*, and *Rejected*. These labels are part of the scientific record; they separate programme-level conjecture from results.

1.3 Evidence Chain

Every substantive assertion should be traceable through:

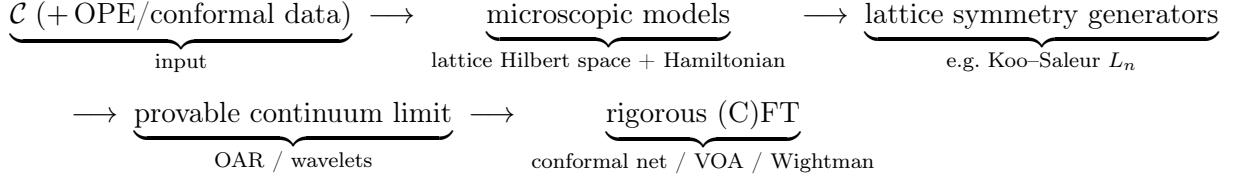
1. a local source manifest under `references/` (the kept papers are the ground truth),
2. a convention entry in `CONVENTIONS.md`,
3. a derivation, script, test, or run bundle when computation is involved,
4. an index row in `INDEX.md`, and
5. a report shard in this lab book.

Missing links are allowed during exploration only when they are explicit.

2 Programme Map

This shard is a map of questions, not a list of results. The north star (§1 / CA-00) factors into a pipeline of stages; each stage is an open problem with its own conventions, sources, and checkable invariants.

2.1 The Pipeline



2.2 Stage 1 — Category Input

- Which categories are in initial scope: multiplicity-free unitary fusion categories (Fibonacci, Ising/TL), then modular tensor categories?
- What auxiliary data beyond $\mathcal{C}$ is genuinely required (OPE coefficients, central charge, conformal weights), and for which target CFTs?
- Which gauge and indexing conventions are fixed before any F/R-symbol is written (see `CONVENTIONS.md`)?

2.3 Stage 2 — Microscopic Models

- Which family realises $\mathcal{C}$ on the lattice: anyon chains, string-nets, or another construction — and how is the family indexed by scale?
- What is the lattice Hilbert space (fusion-tree basis) and the Hamiltonian, and which invariants pin them (fusion-rule consistency, pentagon, projector idempotence)?

2.4 Stage 3 — Lattice Symmetry Generators

- Which lattice operators approximate the continuum symmetry generators (Koo-Saleur lattice Virasoro L_n , $\bar{L}_n$, and kin)?
- What convergence statement is the target, and what is checkable at finite size (commutator algebra, central charge extraction)?

2.5 Stage 4 — Provable Continuum Limit

- Which rigorous renormalisation framework carries the limit (operator- algebraic renormalisation, wavelet / multi-scale entanglement)?
- What must be proved: existence of the limit, the limiting Hilbert space, and convergence of the symmetry generators to a projective unitary representation?

2.6 Stage 5 — Rigorous (C)FT Target

- What is the target object: a conformal net, a vertex operator algebra, a Wightman theory, or an Osterwalder–Schrader theory?
- What is the fidelity statement: that the target’s superselection / representation category recovers $\mathcal{C}$?

2.7 First Decisions

The first technical session should not start with formulas. It should register the sources for one worked input category (Fibonacci or Ising/TL), fix the minimum conventions (F/R-symbol gauge, vacuum index, fusion-tree ordering), and identify the smallest invariant that can be checked end-to-end on one stage of the pipeline.

3 Many-Particle Hilbert Spaces: the AND/OR/Vacuum Grammar

3.1 Sources and Dependencies

This is a pre-topic foundation for the programme (see CA-01, the programme map): how to build the Hilbert space of an indefinite number of particles of arbitrary types. The guiding idea is that two logical connectives map cleanly onto two Hilbert-space constructions, and the no-particle system is the complex line.

This shard treats the *finite* grammar and its semantics; the rigorous *indefinite*-particle (Hilbert-completed) case is CA-03. It harmonises with, and re-proves nothing from, the finite “AND/OR many-particle grammar” developed by the author in the sibling project (notation $\mathcal{V}$, $\mathcal{V}_n$, AND = $\otimes$, OR = $\oplus$, $\mathbf{1} = \mathbb{C}$); see that project’s shard AQM-70. Statistics (Bose / Fermi / para) are deferred throughout.

3.2 The Three Primitives

Let particle *types* be (separable, complex) Hilbert spaces; a type’s space is the state space of one particle of that type. Three primitives generate composite systems:

- **OR = direct sum** $\oplus$. “Either a particle of type 1 or of type 2” is $\mathcal{H}_1 \oplus \mathcal{H}_2$.
- **AND = tensor product** $\otimes$. “A particle of type 1 *and* a particle of type 2” is $\mathcal{H}_1 \otimes \mathcal{H}_2$.
- **vacuum** = $\mathbb{C}$. The no-particle system is the complex line.

Crucially, this grammar fixes *indefinite particle number*; exchange symmetry (statistics) is a *separate, later* step, realised as a quotient or subspace (CA-03, §“Symmetries come later”). Keeping the two apart is the whole point.

3.3 Two Monoidal Structures and Their Units

On the category Hilb of complex Hilbert spaces (with the obvious unitary coherence isomorphisms) there are two symmetric monoidal structures:

$$(\text{Hilb}, \oplus, \{0\}) \quad \text{and} \quad (\text{Hilb}, \otimes, \mathbb{C}),$$

with the two units *distinct*:

$$A \oplus \{0\} \cong A \quad (\oplus\text{-unit is the zero space } \{0\}),$$

$$A \otimes \mathbb{C} \cong A \quad (\otimes\text{-unit is the vacuum } \mathbb{C}).$$

The two are linked by a *distributivity* natural isomorphism, which on Hilbert spaces is *unitary*:

$$\delta : A \otimes (B \oplus C) \xrightarrow{\cong} (A \otimes B) \oplus (A \otimes C), \quad A \otimes \{0\} \cong \{0\}.$$

Together $(\text{Hilb}, \oplus, \{0\}, \otimes, \mathbb{C})$ is a symmetric *rig* (bimonoidal) category. Its coherence — that all diagrams built from associators, unitors, symmetries, and δ commute — is standard (Laplaza; Kelly) and is *not* re-proved here; downstream we use only the elementary, manifestly unitary associativity/unitor isomorphisms and the degree-0 identification $\mathbb{C} \otimes A \cong A$.

Vacuum = $\mathbb{C}$, **not** $\{0\}$. The vacuum is the $\otimes$ -unit $\mathbb{C}$ (the empty tensor product, one particle's worth of “nothing”), a *one-dimensional* space carrying the unit vector Ω . It is *not* the $\oplus$ -unit $\{0\}$ (the empty direct sum, the impossible system). Empty $\otimes$ is $\mathbb{C}$; empty $\oplus$ is $\{0\}$. Conflating them is the first pitfall this grammar guards against (cf. CONVENTIONS.md (a), which fixes the vacuum *object* as $\mathbb{C}$).

3.4 The Grammar and its Carrier

Fix a set $\{K_x\}_{x \in X}$ of single-particle type spaces (the *atoms*). Grammar expressions are

$$E ::= 0 \mid \mathbf{1} \mid K \mid \text{AND}(E, F) \mid \text{OR}(E, F),$$

where 0 is the zero atom, $\mathbf{1}$ is the vacuum atom ($\mathbf{1} = \mathbb{C}$), and K ranges over the atoms. The *carrier* $\mathcal{V}(E)$ is defined recursively:

$$\begin{aligned} \mathcal{V}(0) &= \{0\}, & \mathcal{V}(\mathbf{1}) &= \mathbb{C}, & \mathcal{V}(K) &= K, \\ \mathcal{V}(\text{AND}(E, F)) &= \mathcal{V}(E) \otimes \mathcal{V}(F), & \mathcal{V}(\text{OR}(E, F)) &= \mathcal{V}(E) \oplus \mathcal{V}(F). \end{aligned}$$

Thus AND is independent coexistence and OR is alternatives. Parenthesisation and order are part of the written expression; the rig coherence above guarantees that re-bracketing or re-ordering a *fixed* expression changes $\mathcal{V}(E)$ only by a canonical unitary isomorphism, so $\mathcal{V}$ is well defined up to canonical unitary isomorphism. (Morally $\mathcal{V}$ is the evaluation of the free symmetric rig category on $\{K_x\}$ into Hilb; we use this only as motivation, not as a load-bearing theorem.)

3.5 Particle-Number Grading

Declare each atom K to be *one* particle and $\mathbf{1}$ to be *zero* particles. The carrier then carries a particle-number grading $\mathcal{V}_n$:

$$\begin{aligned} \mathcal{V}_0(\mathbf{1}) &= \mathbb{C}, \quad \mathcal{V}_n(\mathbf{1}) = \{0\} \ (n \neq 0), \quad \mathcal{V}_1(K) = K, \quad \mathcal{V}_n(K) = \{0\} \ (n \neq 1), \quad \mathcal{V}_n(0) = \{0\}, \\ \mathcal{V}_n(\text{OR}(E, F)) &= \mathcal{V}_n(E) \oplus \mathcal{V}_n(F), \quad \mathcal{V}_n(\text{AND}(E, F)) = \bigoplus_{i+j=n} \mathcal{V}_i(E) \otimes \mathcal{V}_j(F). \end{aligned}$$

For a *finite* expression the parse tree is finite, so only finitely many sectors are nonzero and $\mathcal{V}(E) = \bigoplus_{n \geq 0} \mathcal{V}_n(E)$ is a *finite* direct sum. The integer n is a grammar count of one-particle atoms; it is not a dynamics, a statistics rule, or (yet) an indefinite-particle space.

Worked atoms. A one-particle “menu” over a finite type set X is $P = \text{OR}_{x \in X} K_x$, with $\mathcal{V}(P) = \bigoplus_{x \in X} K_x$ (the infinite, ℓ^2 -completed menu is CA-03). The expression $\mathbf{1} \text{ OR } K$ has carrier $\mathbb{C} \oplus K$ — a “maybe” particle (either zero or one particle of type K), as in the sibling shard’s $\mathbb{C} \oplus \mathbb{C}^d$. Tensoring N “maybe” atoms, $\text{AND}_{j=1}^N (\mathbf{1} \text{ OR } K)$, has carrier $(\mathbb{C} \oplus K)^{\otimes N}$, whose grade- n sector (by distributivity) is $\bigoplus_{|S|=n} \bigotimes_{x \in S} K$ over n -subsets of the N slots — the finite “occupation” picture.

3.6 Lamport Dependency Skeleton

D1 : atoms $\{K_x\}_{x \in X}$ (single-particle Hilbert spaces); $\mathbf{1} = \mathbb{C}$ (vacuum), $0 = \{0\}$.

D2 : $E ::= 0 \mid \mathbf{1} \mid K \mid \text{AND}(E, F) \mid \text{OR}(E, F)$.

D3 : $\mathcal{V} : \mathcal{V}(0) = \{0\}, \mathcal{V}(\mathbf{1}) = \mathbb{C}, \mathcal{V}(\text{AND}) = \otimes, \mathcal{V}(\text{OR}) = \oplus$.

D4 : $\mathcal{V}_0(\mathbf{1}) = \mathbb{C}, \mathcal{V}_1(K) = K; \mathcal{V}_n(\text{OR}) = \oplus, \mathcal{V}_n(\text{AND}) = \bigoplus_{i+j=n} \mathcal{V}_i \otimes \mathcal{V}_j$.

D5 : (Hilb, $\oplus, \{0\}, \otimes, \mathbb{C}$) a symmetric rig category, with

$$\delta : A \otimes (B \oplus C) \xrightarrow{\cong} (A \otimes B) \oplus (A \otimes C) \text{ unitary and natural.}$$

T1 : (D5) coherence (Laplaza; Kelly) $\Rightarrow \mathcal{V}$ well defined up to canonical unitary iso.

T2 : (D4) finite $E : \mathcal{V}(E) = \bigoplus_{n \geq 0} \mathcal{V}_n(E)$ a finite sum (finite particle number).

3.7 Scope and Deferral

Two extensions are deliberately out of this shard:

- **Indefinite particle number.** A genuine superposition over *unboundedly* many particle numbers has infinite support and does not live in the finite direct sum above; it requires the Hilbert (ℓ^2) completion. That is the content of CA-03, and is the new rigorous part of this pre-topic.
- **Statistics.** AND and OR choose no exchange symmetry. Bosonic / fermionic / para sectors arise later, by acting with the permutation group on tensor slots and passing to an invariant subspace or quotient (CA-03). This grammar only sets up the carrier those symmetries act on.

Pitfalls. (i) The $\oplus$ -unit is $\{0\}$; the $\otimes$ -unit (the vacuum) is $\mathbb{C}$ — distinct objects. (ii) Distributivity is an *isomorphism*, not an equality; all sector identities below hold up to canonical unitary iso. (iii) Everything here is *finite*; do not silently read the grading as an indefinite-particle space.

4 Indefinite Particle Number: the Full Fock Completion

4.1 Sources and Dependencies

This shard completes CA-02’s finite AND/OR/vacuum grammar to a genuine *indefinite*-particle Hilbert space. It depends on CA-02 (the two monoidal structures, the units $\{0\}$ and $\mathbb{C}$, the carrier $\mathcal{V}$ and grading $\mathcal{V}_n$). The named Fock construction is anchored locally:

The sibling project’s finite grammar (AQM-70) is recovered as the finite-truncation of the object built here.

4.2 Two Direct Sums — the Load-Bearing Distinction

Let $\{W_n\}_{n \geq 0}$ be a countable family of Hilbert spaces.

- The **algebraic direct sum** $\bigoplus_n^{\text{alg}} W_n$ consists of finite-support sequences (w_n) ($w_n \in W_n$, $w_n = 0$ for all but finitely many n). It is a pre-Hilbert space, generally *incomplete*.
- The **Hilbert (ℓ^2) direct sum** $\widehat{\bigoplus}_n W_n$ consists of sequences with $\sum_n \|w_n\|_{W_n}^2 < \infty$, with inner product $\langle (w_n), (v_n) \rangle = \sum_n \langle w_n, v_n \rangle_{W_n}$ (absolutely convergent by Cauchy–Schwarz). It is *complete*, hence a Hilbert space.

$\bigoplus_n^{\text{alg}} W_n$ is a *dense* subspace of $\widehat{\bigoplus}_n W_n$, and the latter is its completion (standard functional analysis).

Why indefinite particle number forces $\widehat{\bigoplus}$. A state with nonzero amplitude in *unboundedly* many particle-number sectors (a genuine superposition over indefinitely many particles) has infinite support: it lies in $\widehat{\bigoplus}$ but not in $\bigoplus^{\text{alg}}$. “Indefinite particle number” $\Leftrightarrow$ “not confined to finitely many sectors” $\Rightarrow$ the ℓ^2 completion is *required*. This is exactly what CA-02’s finite grammar lacks.

4.3 Completed Tensor Powers

For Hilbert spaces H, K , the *completed* (Hilbert) tensor product $H \widehat{\otimes} K$ is the completion of the algebraic tensor product under the inner product determined by $\langle a \otimes b, c \otimes d \rangle = \langle a, c \rangle_H \langle b, d \rangle_K$ (well defined and positive definite). For finite-dimensional factors $\widehat{\otimes} = \otimes$; the completion bites only for infinite-dimensional one-particle spaces. Define the *completed tensor powers*

$$H^{\widehat{\otimes} 0} := \mathbb{C} \text{ (the vacuum sector),} \quad H^{\widehat{\otimes} n} := \underbrace{H \widehat{\otimes} \cdots \widehat{\otimes} H}_n,$$

the bracketing immaterial up to the canonical unitary associativity isomorphism.

4.4 The Full Fock Space

Definition. For a one-particle Hilbert space H , the *full Fock space* is

$$\mathfrak{F}(H) := \widehat{\bigoplus_{n \geq 0} H^{\widehat{\otimes} n}}, \quad H^{\widehat{\otimes} 0} = \mathbb{C},$$

i.e. $\psi = (\psi_n)_{n \geq 0}$ with $\psi_n \in H^{\widehat{\otimes} n}$ and $\|\psi\|^2 = \sum_n \|\psi_n\|^2 < \infty$. The unit vector $\Omega = (1, 0, 0, \dots)$ in the degree-0 sector $\mathbb{C}$ is the *vacuum*. This is the *distinguishable* Fock space (the author’s term in the local source 1910.10619); in operator algebra it is also called the *free* or *Boltzmann* Fock

space (standard terminology, not anchored in a local source), namely the Hilbert completion of the tensor algebra $T(H) = \bigoplus_n H^{\otimes n}$. It is *not* the symmetric (bosonic) or antisymmetric (fermionic) Fock space; see “Exchange sectors come later” below.

Compiler status. In the compiler grammar this is not an additional primitive constructor. It is the Hilbert completion of the ordinary tensor-power expressions $\mathbf{1}, H, H^{\widehat{\otimes} 2}, \dots$: a countable OR over all finite AND powers. If a one-particle symmetry representation is present, CA-05 makes this same object a graded direct-sum representation.

Theorem (Fock is a Hilbert space). $\mathfrak{F}(H)$ is a Hilbert space; (i) the algebraic direct sum $\bigoplus_n^{\text{alg}} H^{\widehat{\otimes} n}$ (the finite-particle vectors) is *dense*; (ii) distinct- n sectors are *mutually orthogonal*, $\langle \psi, \varphi \rangle = \sum_n \langle \psi_n, \varphi_n \rangle$, so $H^{\widehat{\otimes} m} \perp H^{\widehat{\otimes} n}$ for $m \neq n$; (iii) on the degree-0 sector the inner product is the standard one on $\mathbb{C}$ and Ω is a unit vector. *Proof.* An instance of the ℓ^2 -direct-sum facts above with $W_n = H^{\widehat{\otimes} n}$: completeness, density of the algebraic sum, and summand orthogonality are exactly those statements. $\square$

The $1/N!$ caveat (a convention trap). Schottenloher’s inner product on $D = \bigoplus_N H_N$ weights sector N by $1/N!$; that factor is a *symmetric*-Fock normalisation correcting the $N!$ overcounting of symmetric tensors. The full Fock space above uses the *plain* ℓ^2 sum with *no* combinatorial weight. Do not copy the $1/N!$ into the distinguishable case.

4.5 The Grammar Generates the Indefinite-Particle Space

Let the one-particle menu over a countable type set X be $P = \text{OR}_{x \in X} K_x$, with carrier $\mathcal{V}(P) = \bigoplus_{x \in X} K_x$ (the ℓ^2 direct sum; an ordinary finite direct sum when X is finite). The indefinite-particle space over this menu is $\mathfrak{F}(\mathcal{V}(P))$. Expanding each completed tensor power by distributivity (CA-02’s δ , extended unitarily to the completions) gives the $(n, \text{type-word})$ -graded decomposition

$$\mathcal{V}(P)^{\widehat{\otimes} n} \cong \bigoplus_{w \in X^n} K_w, \quad K_w := K_{x_1} \widehat{\otimes} \cdots \widehat{\otimes} K_{x_n}, \quad \implies \quad \mathfrak{F}(\mathcal{V}(P)) \cong \bigoplus_{n \geq 0} \bigoplus_{w \in X^n} K_w.$$

Each of the n tensor slots independently chooses a type $x_i \in X$ (two slots may choose the same type; nothing is symmetrised). The grading is $\mathcal{V}_n = \bigoplus_{w \in X^n} K_w$ (n = particle number, $w \in X^n$ = ordered type-word) — the ℓ^2 -completed, all- n analogue of CA-02’s finite grade- n sector. The sibling project’s finite expressions ($P_X^{\text{AND } m}$, the truncation $T_{\leq M}$) are precisely the finite-support / finite- M restrictions of this object; that is the exact harmonisation: AQM-70 is the $\bigoplus^{\text{alg}} / \text{finite-}M$ truncation of $\mathfrak{F}$.

4.6 Exchange Sectors Come Later

The full Fock space already carries symmetry representations when H does: internal symmetries act sectorwise by tensor powers (CA-05), and for each n the permutation group S_n acts unitarily on $H^{\widehat{\otimes} n}$ by permuting tensor slots. What is deferred is the *sector selection*. Bosonic and fermionic Fock spaces are the images of the per-sector orthogonal projectors

$$\Gamma_s(H) = P_{\text{sym}} \mathfrak{F}(H) \quad (\text{symmetric / Bose}), \quad \Gamma_a(H) = P_{\text{anti}} \mathfrak{F}(H) \quad (\text{antisymmetric / Fermi}),$$

with para-statistics the other Young-symmetry isotypic components. This is the “impose an equivalence relation under particle exchange” step. We define no projector and prove nothing here:

the construction of these symmetry quotients/subspaces, and the genuinely anyonic case (where standard Fock space is not the right object), are deferred to later shards.

4.7 Lamport Dependency Skeleton

$$\begin{aligned}
D6 : \bigoplus^{\text{alg}}(\text{finite support}) &\subset \widehat{\bigoplus}(\ell^2) : \langle (w_n), (v_n) \rangle = \sum_n \langle w_n, v_n \rangle. \\
D7 : \text{completed tensor power } H^{\widehat{\otimes} n}, H^{\widehat{\otimes} 0} &= \mathbb{C} \text{ (vacuum)}. \\
D8 : \mathfrak{F}(H) := \widehat{\bigoplus}_{n \geq 0} H^{\widehat{\otimes} n} &\text{ (full / distinguishable Fock); } \Omega = (1, 0, \dots). \\
D9 : P = \text{OR}_{x \in X} K_x; \quad \mathfrak{F}(\mathcal{V}(P)) &= \widehat{\bigoplus}_{n \geq 0} \mathcal{V}(P)^{\widehat{\otimes} n}. \\
T3 : (D6, D7) \mathfrak{F}(H) \text{ Hilbert; } \bigoplus^{\text{alg}} &\text{ dense; } H^{\widehat{\otimes} m} \perp H^{\widehat{\otimes} n} \text{ (} m \neq n \text{)}. \\
T4 : \text{indefinite particle no.} \Leftrightarrow \text{infinite support} &\Rightarrow \widehat{\bigoplus} \text{ (not } \bigoplus^{\text{alg}} \text{)}. \\
T5 : (D8, \delta) \mathfrak{F}(\mathcal{V}(P)) \cong \widehat{\bigoplus}_{n \geq 0} \widehat{\bigoplus}_{w \in X^n} K_w &\text{ ((} n, \text{type-word) grading)}. \\
T6 : \Gamma_s = P_{\text{sym}} \mathfrak{F}, \Gamma_a = P_{\text{anti}} \mathfrak{F} &\text{ deferred (statistics later).}
\end{aligned}$$

4.8 Scope and Pitfalls

In scope: the indefinite-particle direction, i.e. countable OR (the Fock completion). **Out of scope:** countable AND (an *infinite tensor product*), which is analytically a different object (von Neumann's infinite tensor product, needing a stabilising reference sequence) — not developed here.

Pitfalls: (i) *algebraic* $\bigoplus^{\text{alg}}$ (finite particle number, dense, incomplete) vs. *completed* $\widehat{\bigoplus}$ (indefinite, complete) — the distinction this shard exists to make. (ii) For infinite-dimensional one-particle H , the AND of CA-02 must mean the *completed* $\widehat{\otimes}$, not the algebraic $\otimes$. (iii) The $1/N!$ symmetric-Fock weight must not be imported into the distinguishable case. (iv) All sector identities hold on dense subspaces and extend by continuity; states are ℓ^2 -summable and cross-sector inner products vanish.

5 A Hilbert-Space Compiler Contract

5.1 Status and Local Anchors

Status: Proposal / contract shard. This shard introduces no new mathematical classification theorem. It records a design target for the pre-work sequence around CA-02 and CA-03: a typed grammar of many-particle expressions should be compilable to Hilbert-space data and observable-algebra data with explicit evidence status.

Local anchors are CA-02 for the finite AND/OR/vacuum grammar, CA-03 for the Hilbert direct-sum Fock completion, and `CONVENTIONS.md` entries (f) and (g) for symmetry and sector semantics. The first executable check for this sequence is the finite averaging projector in `test/runtests.jl`, backed by `src/CftAnyons.jl`.

5.2 Mini North Star

Proposal: build a *Hilbert-space compiler*. The user supplies an expression in a typed grammar, together with the atom table and required policies; the compiler returns the Hilbert carrier, its observable algebra policy, its symmetry representation data, and any sector projection or quotient presentation that was requested.

The motivating analogy is “regular expressions for quantum mechanics”: a small syntax has compositional semantics. The analogy is only a prompt. Here the syntax is typed and semantics-bearing; every constructor must say what it does to Hilbert carriers, particle gradings, observable algebras, symmetries, and evidence metadata.

5.3 Input Record

A compiled expression is not just a string. It has an input record:

$$\text{Input} = (\text{Atoms}, E, \text{Completion}, \text{ObsPolicy}, \text{Symmetry}, \text{SectorPolicy}).$$

Atoms. A table of atoms. At minimum, each atom names a Hilbert space K_x . Later entries may include a group representation, a particle degree, a source locator, or category data.

E . A grammar expression built from $0, \mathbf{1}, K_x, \text{AND}, \text{OR}$, and later sector constructors. Fock-style all- n notation is treated as a derived completion of tensor powers, not as a primitive grammar constructor.

Completion. Whether $\oplus, \otimes$ are finite Hilbert operations, Hilbert completions $\widehat{\oplus}, \widehat{\otimes}$, or a finite cutoff. CA-03 fixes the first infinite completion of the derived all-tensor-power construction used here.

ObsPolicy. The observable-algebra choice. There is no silent default: full $\mathcal{B}(H)$, block-preserving, tensor-local, symmetry-invariant, and compressed sector algebras are different outputs.

Symmetry. Optional unitary or projective-unitary representation data, handled according to `CONVENTIONS.md` (f).

SectorPolicy. Optional projection, subspace, or quotient data; see `CONVENTIONS.md` (g).

5.4 Output Record

The intended output is a typed record, not only a Hilbert-space dimension:

$$\text{Compile}(E) = (H_E, \{H_{E,n}\}_{n \geq 0}, \mathcal{A}_E, U_E, P_E, Q_E, \text{Evidence}_E).$$

H_E . The Hilbert carrier. CA-02 and CA-03 define the first carrier semantics.

$\{H_{E,n}\}$. The particle-number grading when the atom table declares particle degrees.

$\mathcal{A}_E$. The selected observable algebra. This slot is mandatory once any observable claim is made.

U_E . The propagated symmetry representation, if symmetry data were supplied.

P_E . The selected sector projection, if a sector constructor was used.

Q_E . The quotient presentation, if one was requested and its closed relation subspace was named.

Evidence_E . Source-local, checked, proposal, question, or rejected evidence tags for the pieces above.

5.5 Observable-Algebra Policies

The carrier alone does not determine the observable algebra relevant to a model. For example, $H \oplus K$ supports at least:

$$\mathcal{B}(H \oplus K), \quad \mathcal{B}(H) \oplus \mathcal{B}(K), \quad \text{off-diagonal creation/annihilation blocks between } H \text{ and } K.$$

All three are legitimate in different contexts; none is forced by the word “OR” alone. Similarly, $H \otimes K$ supports the full algebra $\mathcal{B}(H \otimes K)$ and the tensor-local subalgebra generated by $\mathcal{B}(H) \otimes I$ and $I \otimes \mathcal{B}(K)$.

Compiler rule: no report statement may infer an observable algebra merely from a carrier constructor. The statement must name the observable policy, or remain a carrier-only statement.

5.6 Lamport Dependency Skeleton

D10 : Input = (Atoms, E , Completion, ObsPolicy, Symmetry, SectorPolicy).

D11 : $\text{Compile}(E) = (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \text{Evidence}_E)$.

D12 : ObsPolicy $\in$ {carrier-only, $\mathcal{B}(H)$, block, tensor-local, invariant, compressed}.

T7 : A carrier claim needs H_E ; an observable claim also needs $\mathcal{A}_E$.

T8 : The compiler analogy is a proposal, not a theorem of canonical quantization.

5.7 Boundary

This contract does not yet compile fusion-category labels, anyonic braid statistics, Hamiltonians, continuum limits, or CFT data. It only fixes the pre-work target: future shards should be able to say exactly which input record they compile and which output slots they have actually justified.

6 Symmetry-Decorated Many-Particle Grammar

6.1 Sources and Dependencies

This shard extends CA-02 and CA-03 by decorating atoms with symmetry data. It uses CONVENTIONS.md (f).

6.2 Decorated Atoms

Fix a topological group G . A symmetry-decorated atom is a pair

$$(K_x, U_x), \quad U_x : G \rightarrow U(K_x),$$

where K_x is the atomic Hilbert space and U_x is a unitary representation. If the intended symmetry is projective, the pre-work convention is to choose a central extension

$$1 \rightarrow U(1) \rightarrow E \rightarrow G \rightarrow 1$$

and use an honest unitary representation $U_x : E \rightarrow U(K_x)$. The expression is then compiled over E , with a note that the induced physical symmetry is projective over G .

The vacuum atom is $(\mathbb{C}, \mathbf{1})$, where $\mathbf{1}(g)z = z$. The zero atom carries the unique representation on $\{0\}$.

6.3 Propagation Through OR and AND

If E, F have compiled carriers and representations (H_E, U_E) and (H_F, U_F) for the same group or chosen central extension, define

$$H_{\text{OR}(E,F)} = H_E \oplus H_F, \quad U_{\text{OR}(E,F)}(g) = U_E(g) \oplus U_F(g),$$

and

$$H_{\text{AND}(E,F)} = H_E \hat{\otimes} H_F, \quad U_{\text{AND}(E,F)}(g) = U_E(g) \hat{\otimes} U_F(g).$$

For finite-dimensional factors, $\hat{\otimes} = \otimes$. In infinite-dimensional Hilbert spaces, the completed tensor product is the one already required by CA-03.

These formulas are the Hilbert-space version of the monoidal structure on $\text{Rep}^\dagger(G)$: direct sums are block actions and tensor products are diagonal actions. If two atoms use different groups, different central extensions, or incompatible projective multipliers, the compiler must stop and ask for a common symmetry object before forming a single U_E .

6.4 The Derived Full-Fock Representation

For an ordinary unitary representation $U : G \rightarrow U(H)$, the full-Fock object is itself a graded representation:

$$\mathfrak{F}(U)(g) = \bigoplus_{n \geq 0} U(g)^{\hat{\otimes} n} \quad \text{on} \quad \mathfrak{F}(H) = \widehat{\bigoplus_{n \geq 0} H^{\hat{\otimes} n}},$$

with $U(g)^{\hat{\otimes} 0} = \text{id}_{\mathbb{C}}$. This preserves the particle-number grading. This is not a new grammar constructor: it is the direct sum of the tensor-power representations already produced by OR and AND, completed as in CA-03. In this pre-work shard, no Shale–Stinespring-type implementability theorem is claimed for general canonical field algebras. Implementing a one-particle unitary on a chosen CAR/CCR observable algebra is a separate field-algebra question.

6.5 Intertwiners and Symmetry-Preserving Compilation

A morphism between decorated atoms or compiled expressions is symmetry-preserving only when it is an intertwiner:

$$TU_E(g) = U_F(g)T \quad \text{for all } g.$$

This matters for compiler optimisations. Rebracketing and distributivity isomorphisms used by CA-02 are acceptable in the decorated grammar only when they also intertwine the propagated actions. The standard Hilbert-space associators, unitors, symmetries, and distributivity maps do so for diagonal direct-sum and tensor-product actions.

6.6 Lamport Dependency Skeleton

D13 : decorated atom (K_x, U_x) , $U_x : G \rightarrow U(K_x)$.

D14 : projective symmetry over G is compiled via $U_x : E \rightarrow U(K_x)$ for a named central extension E .

D15 : $U_{E \text{ OR } F} = U_E \oplus U_F$, $U_{E \text{ AND } F} = U_E \hat{\otimes} U_F$.

D16 : $\mathfrak{F}(U) = \widehat{\bigoplus_{n \geq 0} U^{\hat{\otimes} n}}$, $U^{\hat{\otimes} 0} = 1_{\mathbb{C}}$.

T9 : The propagated symmetry preserves the CA-02/CA-03 particle grading.

T10 : A single compiled U_E requires a common group or central extension.

6.7 Boundary

This shard does not yet impose invariance, select charge sectors, form quotient spaces, or define symmetry-invariant observable algebras. Those are sector operations, not mere symmetry propagation, and are treated in CA-06.

7 Sectors, Projections, Subspaces, and Quotients

7.1 Sources and Dependencies

This shard depends on CA-04 and CA-05 and fixes the first sector semantics for the Hilbert-space compiler. It uses CONVENTIONS.md (g).

7.2 The Triangle

Let H be a Hilbert space. The compiler treats a sector first as an orthogonal projection

$$P = P^* = P^2 \in \mathcal{B}(H).$$

It may then expose either of the equivalent Hilbert-space presentations

$$H_P := \text{Ran}(P) \subset H, \quad H/\ker(P) = H/\text{Ran}(1 - P).$$

The map

$$[v] \in H/\ker(P) \mapsto Pv \in \text{Ran}(P)$$

is a well-defined unitary isomorphism when the quotient has the Hilbert quotient norm. Thus the safe slogan is not “symmetry equals quotient” but:

$$\text{projection} \iff \text{closed subspace} \iff \text{quotient by the named orthogonal complement.}$$

7.3 Finite-Group Invariants

Let G be finite and $U : G \rightarrow U(H)$ a unitary representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} U(g).$$

Then P_G is the orthogonal projection onto the invariant subspace

$$H^G = \{v \in H : U(g)v = v \ \forall g \in G\}.$$

Proof. Idempotence follows by reindexing:

$$P_G^2 = \frac{1}{|G|^2} \sum_{g,h} U(gh) = \frac{1}{|G|} \sum_k U(k) = P_G.$$

Self-adjointness follows from unitarity and $g \mapsto g^{-1}$:

$$P_G^* = \frac{1}{|G|} \sum_g U(g)^* = \frac{1}{|G|} \sum_g U(g^{-1}) = P_G.$$

For every k , $U(k)P_G = P_G$, so $\text{Ran}(P_G) \subset H^G$. If $v \in H^G$, then $P_G v = v$, so $H^G \subset \text{Ran}(P_G)$. $\square$

7.4 Coinvariant Quotient

For the same finite action, let

$$R_G = \text{span}\{U(g)v - v : g \in G, v \in H\}.$$

Then $R_G \subseteq \ker(P_G)$ because $P_GU(g) = P_G$. Conversely, if $w \in \ker(P_G)$, then

$$w = \frac{1}{|G|} \sum_{g \in G} (w - U(g)w) \in R_G.$$

Therefore $R_G = \ker(P_G) = (H^G)^\perp$, and the coinvariant quotient

$$H_G := H/R_G$$

is canonically unitarily isomorphic to the invariant subspace H^G . For compact non-finite groups the analogous statement requires a Haar-integral and analytic hypotheses; this shard records only the finite checked case.

7.5 Observable-Algebra Distinctions

There are three different symmetry-related algebras that the compiler must not identify silently:

$$\mathcal{B}(H), \quad \mathcal{B}(H)^G := \{A : U(g)AU(g)^* = A \ \forall g\}, \quad P_G\mathcal{B}(H)P_G \cong \mathcal{B}(H^G).$$

The first is the full algebra before imposing symmetry. The second is the algebra of invariant observables on the original carrier. The third is the compressed algebra acting on the invariant sector. A model may choose one of these, but the choice is an observable policy in the sense of CA-04, not a consequence of the word “symmetry” alone.

7.6 Checked Example

The current Julia check uses $G = C_2$ acting on $\mathbb{C}^2$ by the swap matrix S . The averaging projection is

$$P = \frac{1}{2}(I + S) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

The test checks $P^* = P = P^2$, checks $I - P$ is the complementary orthogonal projection, verifies $P(I - P) = 0$, and verifies that $(1, 1)$ is fixed while $(1, -1)$ is killed. This is the first executable invariant for the sector compiler.

7.7 Lamport Dependency Skeleton

D17 : sector projection $P = P^* = P^2$; $H_P = \text{Ran}(P)$.

D18 : $H/\ker(P) \xrightarrow{[v] \mapsto Pv} \text{Ran}(P)$.

D19 : $P_G = |G|^{-1} \sum_g U(g)$ (G finite).

D20 : $R_G = \text{span}\{U(g)v - v\}$.

L1 : $P_G = P_G^* = P_G^2$, $\text{Ran}(P_G) = H^G$.

L2 : $R_G = \ker(P_G) = (H^G)^\perp$, $H/R_G \cong H^G$.

T11 : Projection, subspace, and quotient are equivalent only after the relation subspace is named.

T12 : $\mathcal{B}(H)^G$ and $P_G\mathcal{B}(H)P_G$ are different observable policies.

7.8 Boundary

The finite averaging theorem is not a statement about spontaneous symmetry breaking, gauge constraints, BRST cohomology, braid statistics, or superselection sectors in a continuum CFT. It is the first Hilbert-space sector rule needed by the compiler.

8 Exchange Symmetry and Statistics as Sector Selection

8.1 Sources and Dependencies

This shard discharges part of the deferral in CA-03: ordinary Bose/Fermi statistics are treated as choices of irreducible exchange-symmetry sectors on the full distinguishable Fock carrier. It depends on CA-05 for internal symmetry propagation and CA-06 for projection/subspace/quotient semantics.

8.2 Two Symmetry Layers

There are two distinct actions on many-particle carriers:

$$\text{internal symmetry } G, \quad \text{exchange symmetry } S_n.$$

If $U : G \rightarrow U(H)$ is an internal one-particle representation, then on the n -particle distinguishable sector $H^{\widehat{\otimes} n}$ it acts diagonally:

$$U_n(g) = U(g)^{\widehat{\otimes} n}.$$

The permutation group S_n acts by permuting tensor slots:

$$\Pi_n(\sigma)(v_1 \otimes \cdots \otimes v_n) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)}.$$

The two actions commute:

$$\Pi_n(\sigma)U_n(g) = U_n(g)\Pi_n(\sigma).$$

Thus internal charge sectors and exchange-statistics sectors can be imposed in either order in the ordinary tensor-slot setting.

8.3 Irrep-Labeled Exchange Projectors

The representation $\Pi_n : S_n \rightarrow U(H^{\widehat{\otimes} n})$ is the primary datum. Sector projections are supplied by irreducible representation data of S_n , not by specialised grammar constructors. For a one-dimensional unitary character $\chi : S_n \rightarrow U(1)$, CA-06 applied to the twisted representation $\bar{\chi}\Pi_n$ gives

$$P_{\chi,n} = \frac{1}{n!} \sum_{\sigma \in S_n} \overline{\chi(\sigma)} \Pi_n(\sigma),$$

whose range is the χ -sector

$$\{w \in H^{\widehat{\otimes} n} : \Pi_n(\sigma)w = \chi(\sigma)w \ \forall \sigma \in S_n\}.$$

For the trivial character 1, this is the symmetric projector:

$$P_{s,n} = \frac{1}{n!} \sum_{\sigma \in S_n} \Pi_n(\sigma),$$

and for the sign character sgn , this is the antisymmetric projector:

$$P_{a,n} = \frac{1}{n!} \sum_{\sigma \in S_n} \text{sgn}(\sigma) \Pi_n(\sigma).$$

Their ranges are the symmetric and antisymmetric tensor powers:

$$\text{Ran}(P_{s,n}) = H^{\widehat{\otimes}_s n}, \quad \text{Ran}(P_{a,n}) = \bigwedge^n H.$$

The corresponding Fock sectors are

$$\Gamma_s(H) = \widehat{\bigoplus_{n \geq 0} \text{Ran}(P_{s,n})}, \quad \Gamma_a(H) = \widehat{\bigoplus_{n \geq 0} \text{Ran}(P_{a,n})}.$$

Thus “Bose” and “Fermi” name the trivial- and sign-irrep choices for ordinary exchange symmetry. The compiler should record the selected irrep/character and the resulting projection, not a special Bose/Fermi syntax node.

8.4 Quotient Presentations

For a one-dimensional character χ , the sector can also be presented as a quotient of the full tensor algebra by the closed linear span of relations

$$\Pi_n(\sigma)v - \chi(\sigma)v.$$

The trivial character recovers symmetric relations; the sign character recovers antisymmetric signed relations. In the compiler, these quotient presentations are secondary: the primary sector object is the orthogonal projector, because it fixes the inherited Hilbert structure and the compressed observable algebra.

8.5 Para and Anyonic Boundaries

Higher-dimensional irreducible S_n -sectors require central idempotents or matrix-block projectors for the group algebra of S_n . This is exactly the same representation/irrep principle, but this shard records it as a proposal rather than a completed construction; a later shard must source the convention and projector normalization before using it.

Genuinely anyonic exchange is not an S_n -projection problem on ordinary tensor slots. It involves braid-group or categorical data and, in this project, must be handled together with the fusion-category conventions for associators, braiding, and F/R -symbols. No braid-statistics projector is defined here.

8.6 Lamport Dependency Skeleton

- $D21 : U_n(g) = U(g)^{\widehat{\otimes} n}$ (internal symmetry on $H^{\widehat{\otimes} n}$).
- $D22 : \Pi_n : S_n \rightarrow U(H^{\widehat{\otimes} n})$ permutes tensor slots.
- $D23 : P_{\chi,n} = n!^{-1} \sum_{\sigma} \overline{\chi(\sigma)} \Pi_n(\sigma)$ (χ a one-dimensional S_n character).
- $L3 : \Pi_n(\sigma)U_n(g) = U_n(g)\Pi_n(\sigma)$.
- $L4 : P_{\chi,n}$ is an orthogonal projection by CA-06 finite averaging.
- $T13 : \Gamma_s, \Gamma_a$ are the trivial/sign irrep sectors inside full Fock.
- $T14 : \text{Anyonic exchange remains deferred to categorical braid data.}$

8.7 Compiler Consequence

The compiler now has its first nontrivial sector family: given the exchange representation $\Pi_n : S_n \rightarrow U(H^{\widehat{\otimes} n})$, a selected irrep character χ supplies $P_{\chi,n}$, its range, its quotient relations, and the chosen compressed observable algebra. There is no primitive Bose, Fermi, or Fock grammar node in this account; Fock is the derived graded representation of CA-05, and Bose/Fermi are names for particular irrep choices of exchange symmetry.

9 Scoping the Hilbert-Space Compiler

9.1 Status, Sources, and Dependencies

Status: specification shard. This shard turns CA-04’s contract into a first partial compiler specification. Its scope is the pre-work grammar of CA-02–CA-07: finite Hilbert carriers, derived all-tensor-power completions, unitary/projective symmetries via central extensions, and irrep/projection sectors.

9.2 Compile Judgement

The compiler is a *partial* map. A successful judgement has the form

$$\Gamma; \Pi \vdash E \Downarrow (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \text{Ev}_E),$$

where Γ is the atom table and Π is the policy table. A failed judgement has the form

$$\Gamma; \Pi \vdash E \Uparrow \text{reason}.$$

The failure result is part of the specification: incompatible symmetry data, missing observable policy, or an ill-typed sector constructor must stop compilation.

The atom table entry for x has at least a Hilbert space K_x , a degree $d_x \in \mathbb{N}$ (usually 1), evidence metadata, and optionally symmetry data $U_x : G \rightarrow U(K_x)$ or a named projective central extension as in CA-05.

9.3 Surface Language for the First Compiler

The first surface language is deliberately small:

$$E ::= 0 \mid \mathbf{1} \mid x \mid \text{OR}(E, F) \mid \text{AND}(E, F) \mid \text{Sector}_{G,\chi}(E).$$

Here G acts on the compiled carrier and χ is a one-dimensional unitary character; invariants are the case $\chi = 1$. Fock, finite truncation, symmetric sectors, and antisymmetric sectors are derived from OR/AND, representation propagation, and the generic sector operation.

For a one-particle atom x , define tensor-power expressions recursively by

$$x^{[0]} := \mathbf{1}, \quad x^{[m+1]} := \text{AND}(x^{[m]}, x).$$

Then

$$\mathsf{T}_{\leq M}(x) := \text{OR}_{m=0}^M x^{[m]}, \quad \mathsf{T}_{\infty}(x) := \text{Hilbert completion of } \text{OR}_{m \geq 0} x^{[m]}.$$

$\mathsf{T}_{\infty}(x)$ is the usual full/distinguishable Fock space $\widehat{\bigoplus_{m \geq 0} K_x^{\widehat{\otimes} m}}$, but as a derived graded representation.

9.4 Constructor Rules

The carrier, grading, and symmetry rules are recursive:

$$\begin{aligned} H_0 &= \{0\}, & H_1 &= \mathbb{C}, & H_x &= K_x, \\ H_{\text{OR}(E,F)} &= H_E \oplus H_F, & H_{\text{AND}(E,F)} &= H_E \widehat{\otimes} H_F. \end{aligned}$$

The grading rules are CA-02's:

$$H_{\text{OR}(E,F),n} = H_{E,n} \oplus H_{F,n}, \quad H_{\text{AND}(E,F),n} = \bigoplus_{i+j=n} H_{E,i} \hat{\otimes} H_{F,j}.$$

If U_E, U_F are representations of the same group or named central extension, the symmetry rules are

$$U_{\text{OR}}(g) = U_E(g) \oplus U_F(g), \quad U_{\text{AND}}(g) = U_E(g) \hat{\otimes} U_F(g).$$

If the symmetry objects are not the same, the carrier may still compile, but the symmetry output is rejected unless the requested output was explicitly carrier-only.

For the derived tensor-power closures,

$$H_{\text{T}_\infty(x)} = \widehat{\bigoplus_{m \geq 0} K_x^{\otimes m}}, \quad H_{\text{T}_{\leq M}(x)} = \bigoplus_{m=0}^M K_x^{\otimes m}.$$

For finite examples the compiler may report dimensions:

$$\begin{aligned} \dim \text{OR} &= \dim H_E + \dim H_F, \\ \dim \text{AND} &= (\dim H_E)(\dim H_F), \\ \dim \text{T}_{\leq M}(x) &= \sum_{m=0}^M (\dim K_x)^m. \end{aligned}$$

These follow from choosing orthonormal bases: disjoint union gives direct-sum bases, ordered pairs give tensor-product bases, and truncation is a finite sum.

Sector constructors return projection witnesses supplied by representation data. For finite G , a unitary character $\chi : G \rightarrow U(1)$, and representation $U_E : G \rightarrow U(H_E)$, set

$$P_\chi = |G|^{-1} \sum_{g \in G} \overline{\chi(g)} U_E(g), \quad H_{\text{Sector}_{G,\chi}(E)} = \text{Ran}(P_\chi).$$

The quotient relation is generated by $U_E(g)v - \chi(g)v$. The invariant sector is $\chi = 1$. For exchange sectors on $K^{\hat{\otimes} n}$, $G = S_n$ and $U_E = \Pi_n$; the trivial and sign characters give

$$P_{s,n} = n!^{-1} \sum_{\sigma \in S_n} \Pi_n(\sigma), \quad P_{a,n} = n!^{-1} \sum_{\sigma \in S_n} \text{sgn}(\sigma) \Pi_n(\sigma).$$

The observable algebra is never inferred from these formulas. It is the policy slot Π_{obs} : carrier-only, full, block, tensor-local, invariant, or compressed.

9.5 Example 1: A Maybe Qubit

Let $Q = \mathbb{C}^2$ with basis e_0, e_1 . Compile

$$E_1 = \text{OR}(\mathbf{1}, Q).$$

The rule gives

$$H_{E_1} = \mathbb{C} \oplus \mathbb{C}^2, \quad H_{E_1,0} = \mathbb{C}, \quad H_{E_1,1} = \mathbb{C}^2, \quad \dim H_{E_1} = 3.$$

With full observable policy, $\mathcal{A}_{E_1} = \mathcal{B}(\mathbb{C} \oplus \mathbb{C}^2)$. With block policy,

$$\mathcal{A}_{E_1}^{\text{block}} = \mathcal{B}(\mathbb{C}) \oplus \mathcal{B}(\mathbb{C}^2).$$

Thus the same carrier has two different compiled observable outputs.

9.6 Example 2: Two Distinguishable Qubits

Compile $E_2 = \text{AND}(Q, Q)$. The carrier is

$$H_{E_2} = Q \otimes Q, \quad \{e_0 \otimes e_0, e_0 \otimes e_1, e_1 \otimes e_0, e_1 \otimes e_1\} \text{ is a basis,}$$

so $\dim H_{E_2} = 4$, all in degree 2. The tensor-local policy is the subalgebra generated by

$$\mathcal{B}(Q) \otimes I, \quad I \otimes \mathcal{B}(Q),$$

whereas the full policy is $\mathcal{B}(Q \otimes Q)$. The compiler records which one was requested; it does not identify them by syntax alone.

9.7 Example 3: Derived Tensor-Power Closure

Compile the macro-expanded expression

$$E_3 = \mathsf{T}_{\leq 2}(Q) = \text{OR}(\mathbf{1}, \text{OR}(Q, \text{AND}(Q, Q))).$$

The output carrier is

$$H_{E_3} = \mathbb{C} \oplus Q \oplus (Q \otimes Q),$$

with sector dimensions

$$\dim H_0 = 1, \quad \dim H_1 = 2, \quad \dim H_2 = 4, \quad \dim H_{E_3} = 1 + 2 + 4 = 7.$$

If $U : G \rightarrow U(Q)$ is supplied, symmetry propagation gives

$$U_{E_3}(g) = 1_{\mathbb{C}} \oplus U(g) \oplus (U(g) \otimes U(g)).$$

No Bose/Fermi relation has been imposed yet; this is still the distinguishable graded representation.

9.8 Example 4: Two-Qubit Exchange-Irrep Sectors

Let S be the swap on $Q \otimes Q$. Since

$$S(e_0 e_0) = e_0 e_0, \quad S(e_1 e_1) = e_1 e_1, \quad S(e_0 e_1) = e_1 e_0, \quad S(e_1 e_0) = e_0 e_1,$$

where $e_i e_j$ abbreviates $e_i \otimes e_j$, the exchange representation of S_2 has two one-dimensional irrep characters, trivial and sign, supplying projectors with ranges:

$$P_s = \frac{1}{2}(I + S), \quad P_a = \frac{1}{2}(I - S)$$

$$\text{Ran}(P_s) = \text{span}\{e_0 e_0, (e_0 e_1 + e_1 e_0)/\sqrt{2}, e_1 e_1\},$$

$$\text{Ran}(P_a) = \text{span}\{(e_0 e_1 - e_1 e_0)/\sqrt{2}\}.$$

Thus the compiler reports dimensions 3 and 1 for the trivial- and sign-irrep sectors. The Julia test suite checks the projection, orthogonality, trace/dimension, and basis-vector claims for this example.

9.9 Example 5: A Finite Invariant Quotient

Let $C_2 = \{1, s\}$ act on $H = \mathbb{C}^2$ by the swap matrix. Then

$$P_{C_2} = \frac{1}{2}(I + S) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

Its range is $\text{span}\{(1, 1)\}$, and its kernel is $\text{span}\{(1, -1)\}$. The quotient presentation is therefore

$$\mathbb{C}^2 / \text{span}\{(1, -1)\} \cong \text{span}\{(1, 1)\}.$$

This is $\text{Sector}_{C_2,1}(H)$, the trivial-irrep sector of the C_2 representation. The invariant observable algebra on the original carrier consists of the operators commuting with S , while the compressed algebra on the invariant line is one-dimensional. These are different observable policies even in this small example.

9.10 Example 6: A Required Compile Error

Let A, B be atoms with projective symmetries over the same physical group G , but represented in the input by unrelated central extensions $E_A \rightarrow G$ and $E_B \rightarrow G$. The carrier of $\text{OR}(A, B)$ is still $K_A \oplus K_B$. However, a request for one compiled symmetry representation must fail:

$$\Gamma; \Pi \vdash \text{OR}(A, B) \uparrow \text{“no common symmetry group or central extension specified”}.$$

This is not a missing feature; it is the strict semantics required by CA-05 and CONVENTIONS.md (f).

9.11 Lamport Dependency Skeleton

$D24 : \Gamma; \Pi \vdash E \Downarrow (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \text{Ev}_E)$ or $\Gamma; \Pi \vdash E \uparrow \text{reason}$.

$D25 : E ::= 0 \mid \mathbf{1} \mid x \mid \text{OR} \mid \text{AND} \mid \text{Sector}_{G,\chi}$.

$D26 : \mathbb{T}_{\leq M}(x) = \text{OR}_{m=0}^M x^{[m]}, \quad H_{\mathbb{T}_{\leq M}} = \bigoplus_{m=0}^M K_x^{\otimes m}.$

$D27 : P_\chi = |G|^{-1} \sum_g \chi(g) U_E(g), \quad H_{\text{Sector}_{G,\chi}} = \text{Ran}(P_\chi).$

$T15 : \text{Finite dimensions obey sum, product, and } \sum_{m=0}^M d^m \text{ by explicit basis constructions.}$

$T16 : \text{Irrep/character choices supply projection/subspace/quotient witnesses.}$

$T17 : \text{Compiler failure is a specified outcome when requested structure is ill typed or unsupported.}$

9.12 Acceptance Criteria and Boundary

Future implementation work must preserve these acceptance criteria: every constructor fills typed output slots; every nontrivial mathematical output has an evidence tag; every quotient names its relation subspace; every observable claim names its observable policy; and every executable example asserts an invariant, not merely successful execution.

This compiler does not yet parse fusion-category simples, anyonic braid data, Hamiltonians, lattice limits, conformal nets, or VOA data. Those enter only after their source conventions and invariants are fixed. CA-08 is the finite Hilbert-space kernel that later shards must extend without weakening the output record, the evidence discipline, or the failure semantics.

10 Fibonacci Kinematics as a Fusion-Rule Sector

10.1 Status, Sources, and Dependencies

Status: checked kinematic shard. This shard confirms the side conjecture: before braiding, Hamiltonians, or continuum limits, the Fibonacci fusion-path Hilbert spaces are a sector construction from one physical atom τ . The vacuum $\mathbf{1}$ is still present as the tensor unit and as a charge label; it is not a second physical particle atom.

Dependencies: CA-02 for AND/OR/vacuum, CA-06 for projection/subspace semantics, CA-08 for compiler output slots, and CONVENTIONS.md (c) for the left-associated kinematic path basis.

10.2 One Physical Atom, Two Charge Labels

The physical input word is n copies of one anyon type:

$$\tau, \tau, \dots, \tau.$$

The charge labels used to describe intermediate fusion outcomes are

$$L = \{\mathbf{1}, \tau\}.$$

Thus the proposed slogan is correct only with this distinction:

$$\text{one physical atom } \tau \quad + \quad \text{charge labels } \{\mathbf{1}, \tau\} \quad + \quad \text{fusion-rule sector selection}.$$

The vacuum $\mathbf{1}$ is required because it is both the tensor unit and a possible total charge. It is not an additional non-vacuum particle species.

10.3 Ambient Path Space and Admissibility Projector

Use the left-associated path basis fixed in CONVENTIONS.md (c). For n physical τ -atoms and total charge $c \in \{\mathbf{1}, \tau\}$, define the ambient word space

$$W_{n,c} = \ell^2\{(x_0, \dots, x_n) : x_i \in \{\mathbf{1}, \tau\}, x_0 = \mathbf{1}, x_n = c\}.$$

The basis vector $|x_0, \dots, x_n\rangle$ records the cumulative charge after successively fusing the first i copies of τ .

The Fibonacci fusion rules needed here are

$$\mathbf{1} \otimes \tau = \tau, \quad \tau \otimes \mathbf{1} = \tau, \quad \tau \otimes \tau = \mathbf{1} \oplus \tau.$$

Let $N_{ab}^c \in \{0, 1\}$ be these multiplicity-free fusion coefficients. Define the diagonal operator

$$P_{n,c}|x_0, \dots, x_n\rangle = \left(\prod_{i=1}^n N_{x_{i-1}, \tau}^{x_i} \right) |x_0, \dots, x_n\rangle.$$

Since every diagonal entry is 0 or 1, $P_{n,c} = P_{n,c}^* = P_{n,c}^2$. The kinematic Fibonacci Hilbert space is therefore the sector

$$H_{n,c}^{\text{Fib}} := \text{Ran}(P_{n,c}) \subseteq W_{n,c}.$$

This is exactly the CA-06 projection/subspace semantics, but the projector is not supplied by a group character. It is supplied by categorical fusion-rule admissibility.

10.4 Dimension Recurrence

Let

$$a_n = \dim H_{n, \mathbf{1}}^{\text{Fib}}, \quad b_n = \dim H_{n, \tau}^{\text{Fib}}.$$

At $n = 0$, no physical τ -atom has been fused, so

$$a_0 = 1, \quad b_0 = 0.$$

Adding one more τ gives the transition equations

$$a_{n+1} = b_n, \quad b_{n+1} = a_n + b_n.$$

The first equation says total charge $\mathbf{1}$ can only come from $\tau \otimes \tau \rightarrow \mathbf{1}$. The second says total charge τ can come from $\mathbf{1} \otimes \tau \rightarrow \tau$ or from $\tau \otimes \tau \rightarrow \tau$. Hence

$$b_{n+1} = b_n + b_{n-1}, \quad b_0 = 0, \quad b_1 = 1.$$

Thus the total- τ sector dimensions are Fibonacci numbers:

$$0, 1, 1, 2, 3, 5, 8, \dots$$

The Julia test suite checks the pair sequence

$$(a_n, b_n) = (1, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 5), (5, 8), (8, 13)$$

for $0 \leq n \leq 7$, including the recurrence.

10.5 Small Examples

$n = 1$. For total charge τ , the ambient basis is $|\mathbf{1}, \tau\rangle$, and it is admissible. Thus $H_{1, \tau}^{\text{Fib}} \cong \mathbb{C}$.

$n = 2$. The admissible paths are

$$|\mathbf{1}, \tau, \mathbf{1}\rangle \in H_{2, \mathbf{1}}^{\text{Fib}}, \quad |\mathbf{1}, \tau, \tau\rangle \in H_{2, \tau}^{\text{Fib}}.$$

This is the kinematic content of $\tau \otimes \tau = \mathbf{1} \oplus \tau$.

$n = 3$. The total- τ ambient basis has two admissible paths:

$$|\mathbf{1}, \tau, \mathbf{1}, \tau\rangle, \quad |\mathbf{1}, \tau, \tau, \tau\rangle.$$

The forbidden word $|\mathbf{1}, \mathbf{1}, \tau, \tau\rangle$, for example, is killed because $N_{\mathbf{1}, \tau}^{\mathbf{1}} = 0$. Therefore $\dim H_{3, \tau}^{\text{Fib}} = 2$.

10.6 Compiler Consequence

This is the first example where the Hilbert-space compiler wants category data. At the carrier level, it can compile the ambient finite word space $W_{n, c}$ and the diagonal sector projection $P_{n, c}$. But producing $P_{n, c}$ from the expression “ n copies of τ ” requires the fusion coefficients N_{ab}^c . That is already tensor-category input, not ordinary Hilbert-space grammar.

Moreover, the source makes clear that different fusion orders give different fusion-tree bases and are related by F -moves. Therefore a full Fibonacci compiler must eventually output not only

$$H_{n, c}^{\text{Fib}}$$

but also the coherent recoupling data that identifies different bracketings: the F -symbols satisfying the pentagon equations, and later the R -symbols for braiding. The current shard is only kinematic: it builds the admissible sector and counts it.

10.7 Lamport Dependency Skeleton

- $D28 : L = \{\mathbf{1}, \tau\}, \quad \mathbf{1} \otimes \tau = \tau, \quad \tau \otimes \tau = \mathbf{1} \oplus \tau.$
- $D29 : W_{n,c} = \ell^2\{(x_0, \dots, x_n) : x_0 = \mathbf{1}, x_n = c, x_i \in L\}.$
- $D30 : P_{n,c}|x\rangle = \prod_i N_{x_{i-1}, \tau}^{x_i}|x\rangle.$
- $D31 : H_{n,c}^{\text{Fib}} = \text{Ran}(P_{n,c}).$
- $L5 : P_{n,c} = P_{n,c}^* = P_{n,c}^2.$
- $T18 : \dim H_{n,\tau}^{\text{Fib}} \text{ obeys } b_{n+1} = b_n + b_{n-1}, \quad b_0 = 0, \quad b_1 = 1.$
- $T19 : \text{Full Fibonacci compilation needs } F\text{-coherent tensor-category data.}$

10.8 Boundary

This shard does not fix the Fibonacci F -matrix gauge, any R -symbols, braid representations, golden-chain Hamiltonians, or a continuum limit. It only realises the fusion-path Hilbert spaces kinematically as admissibility sectors.